HeedVision

Attention Awareness in Collaborative Immersive Analytics Environments



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Follow along & ask questions

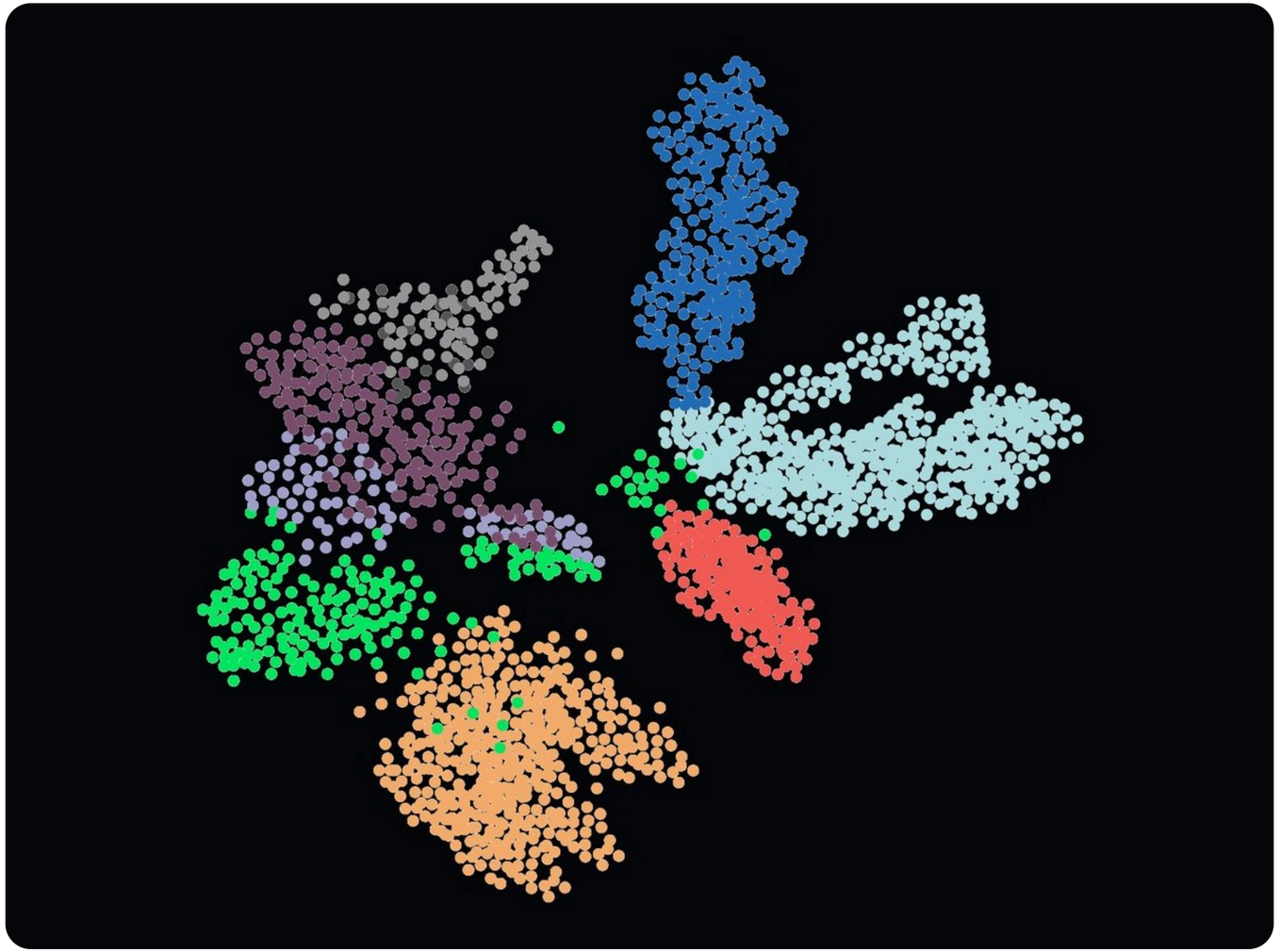
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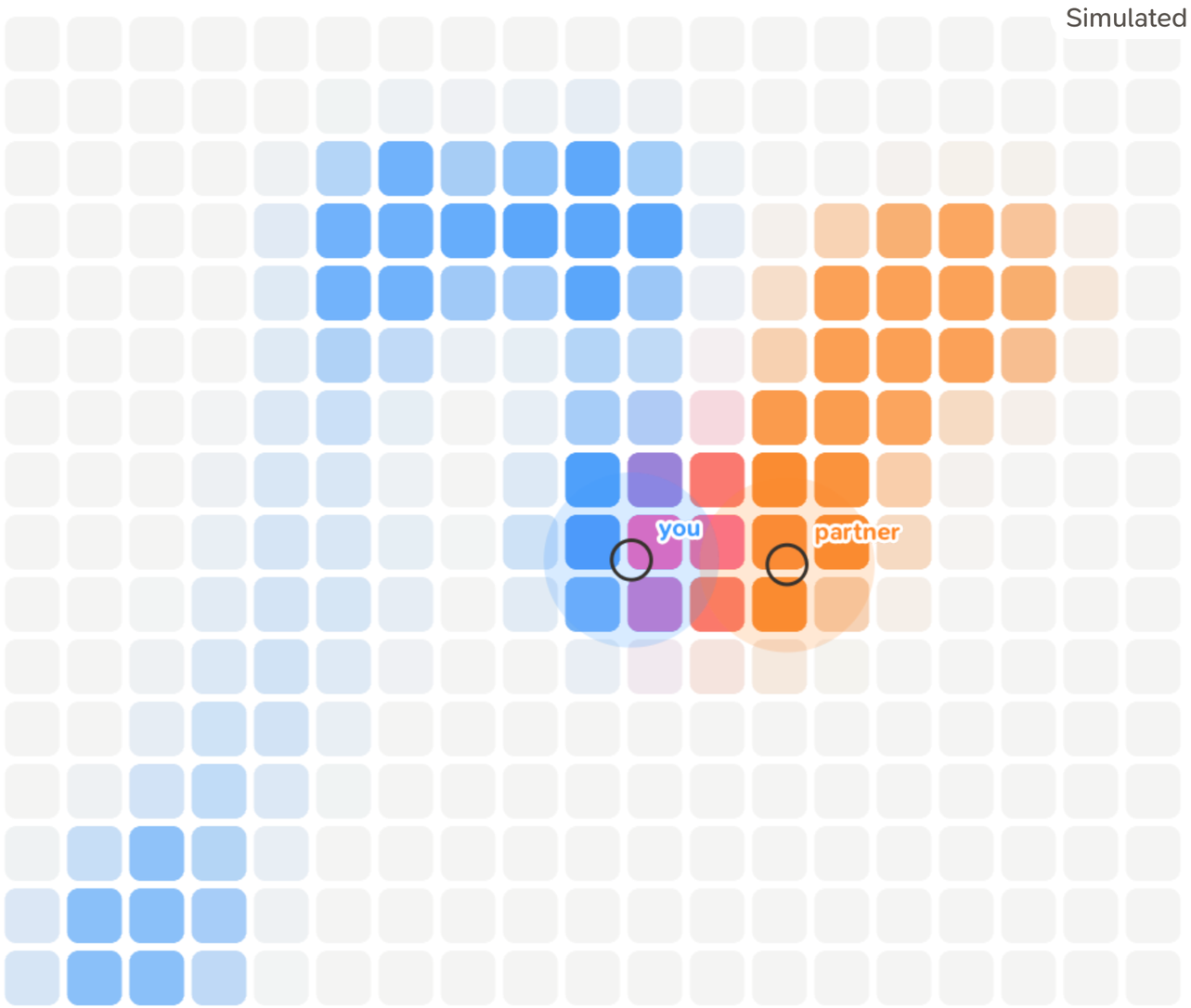
Searching an abstract 3D space

- **No up.** A 3D embedding has no natural orientation.
- **No landmarks.** One cluster looks much like the next.
- **Too much for one person.** So you bring in a colleague.



Example: Legetth et al., CellexalVR, iScience 2021. Image: Single Cell RNA Seq, illustrated UMAP, Ryan Kissinger, courtesy of NIAID (NIH BioArt 767), public domain, background changed.

Who looked where?



- **Group awareness.** The ability to perceive the activities of collaborators in a shared space.
- **Harder in immersive 3D.** Multiple viewpoints, distributed spatial attention, real-time coordination.
- **What if the space remembered?**

1

Idea

From one viewer to many

Collaborative attention-aware visualization



Capture

Distinguish between users while tracking attention simultaneously.

RQ1

How should the attention of multiple viewers on a visualization be captured in real time?



Record

Identity-aware accumulators, with sum, maximum, difference and count.

RQ2

How should the attention of multiple viewers on a visualization be recorded over time?



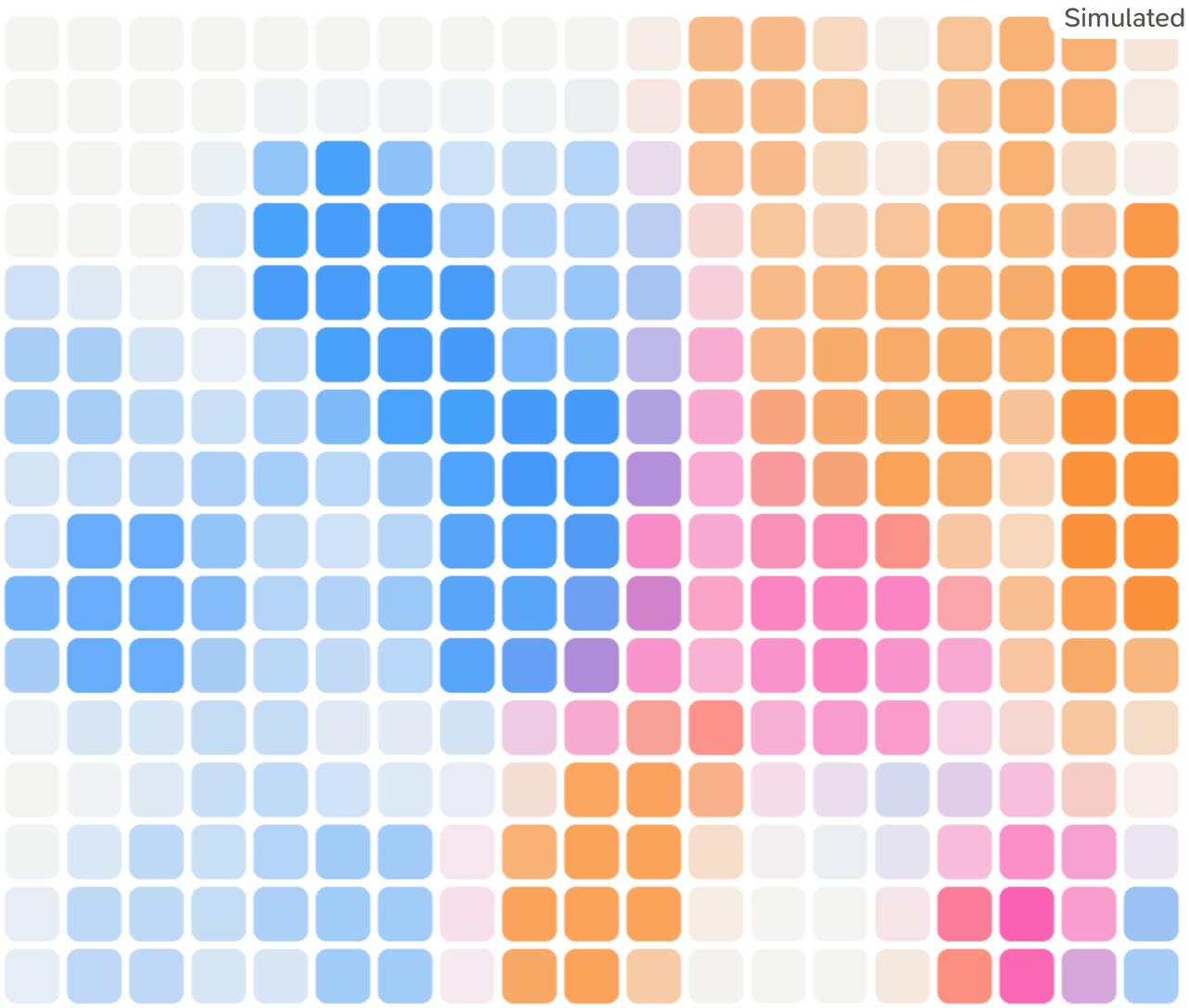
Revisualize

Encode both attention intensity and user identity.

RQ3

How should the attention of multiple viewers on a visualization be displayed?

Record: one session, five views



- **Count.** How many people looked.
- **Sum.** Total coverage.
- **Maximum.** The strongest focus.
- **Difference.** The division of labor.
- **Who.** Blue you, orange partner, pink both.

Display: a design space

Embedded in the vis

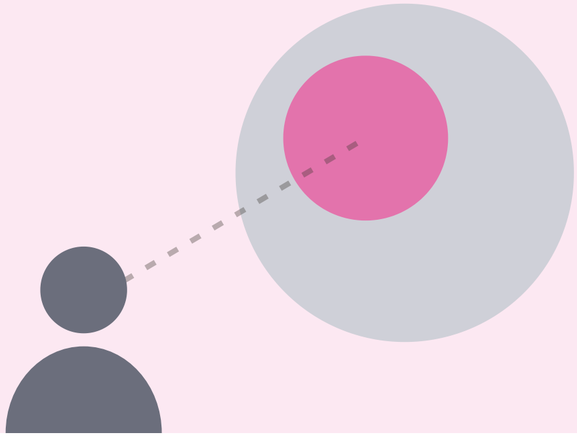
Separate view

Triggering

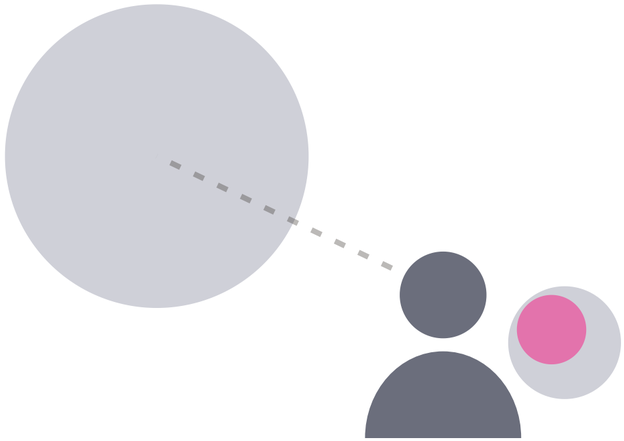
World space

World · embedded

HeedVision

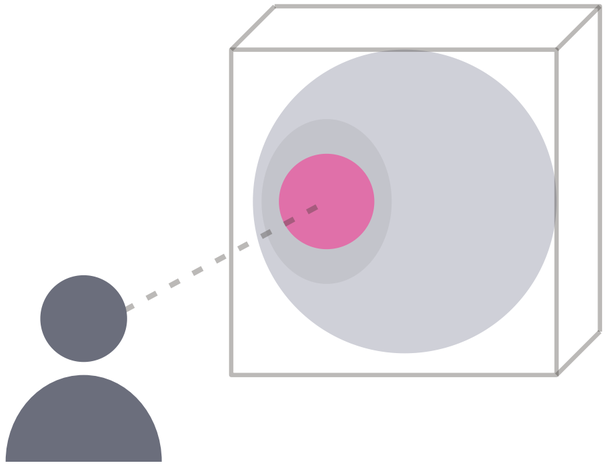


World · separated

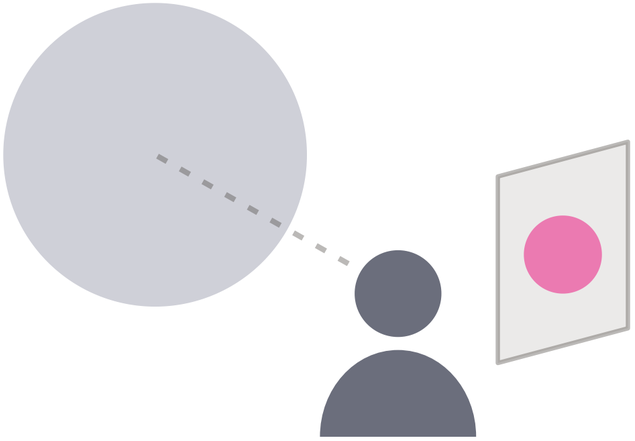


Camera space

Camera · embedded



Camera · separated



Always on

risks Midas touch

Explicit

you ask to see it

Implicit

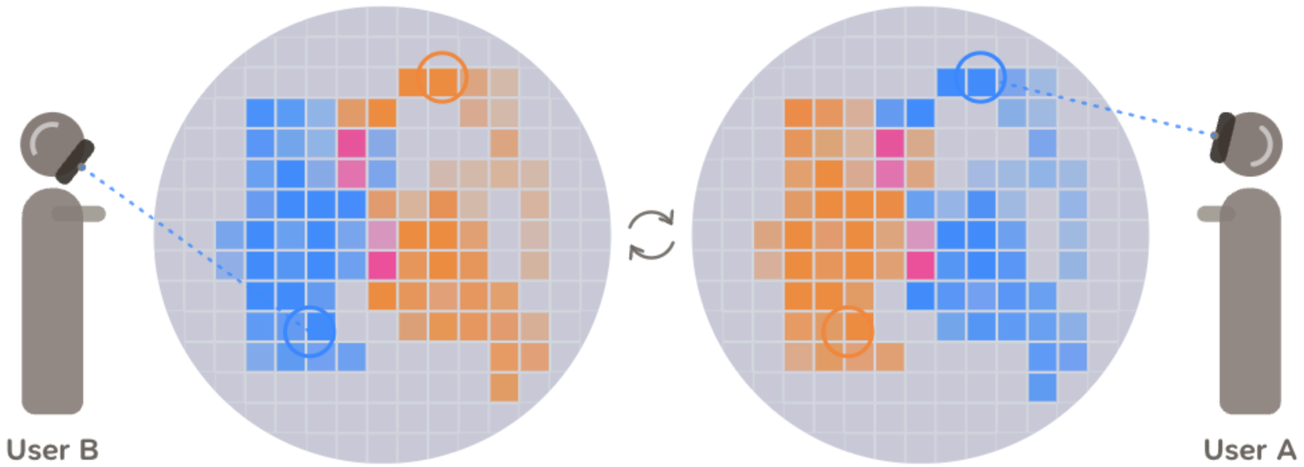
the system decides

2

System

One corner built and evaluated

HeedVision: colour per user

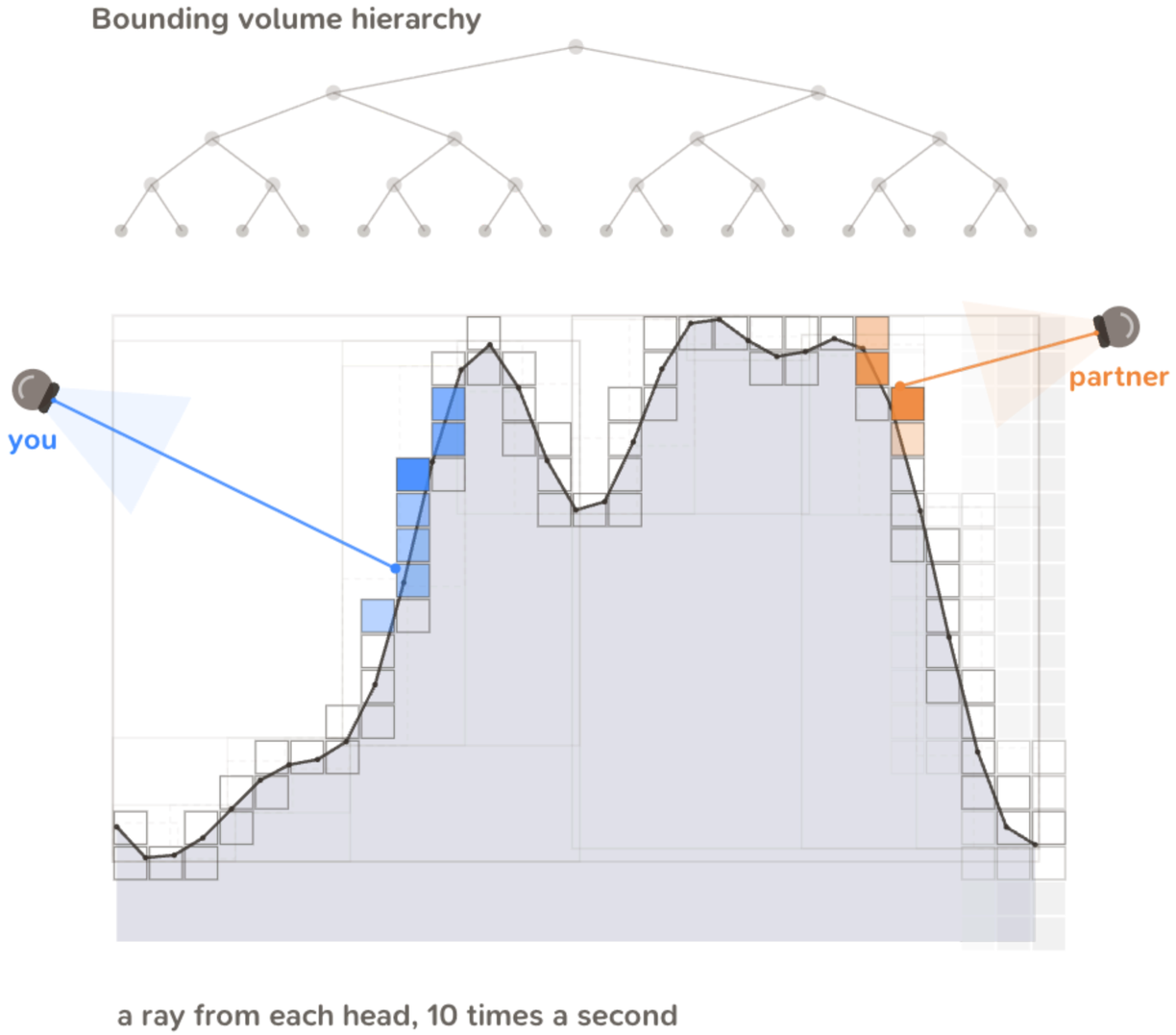


Each headset shows a copy of one shared scene

Trail shown

- **WebXR.** React Three Fiber, on headsets like the Meta Quest 3.
- **Explicit.** A controller button shows or hides attention.
- **Per headset.** Blue is you, orange your partner, pink both.
- **Temporal decay.** Attention fades when you look away.

Voxels collect the attention



- **Wrap the chart.** A bounding volume hierarchy around its geometry.
- **Keep voxels that touch it.** Boxes rule out empty cells.
- **A ray from each head, at 10 Hz.** It hits the nearest voxel, which gains that person's attention.

Display: a design space

Embedded in the vis

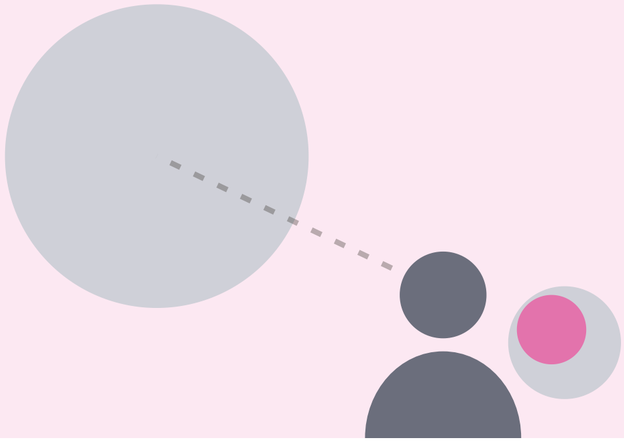
Separate view

Triggering

World space

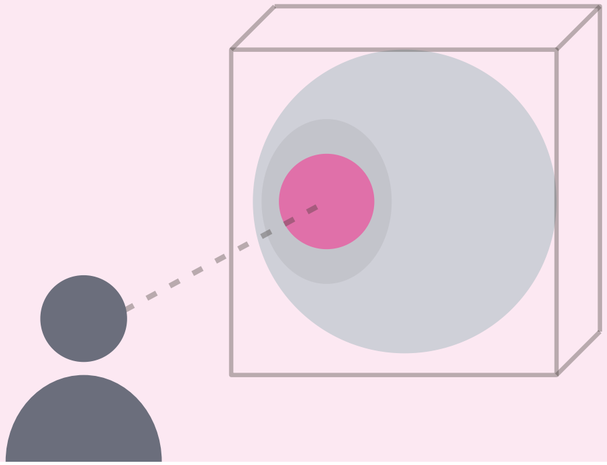
World · embedded

World · separated — Prototype

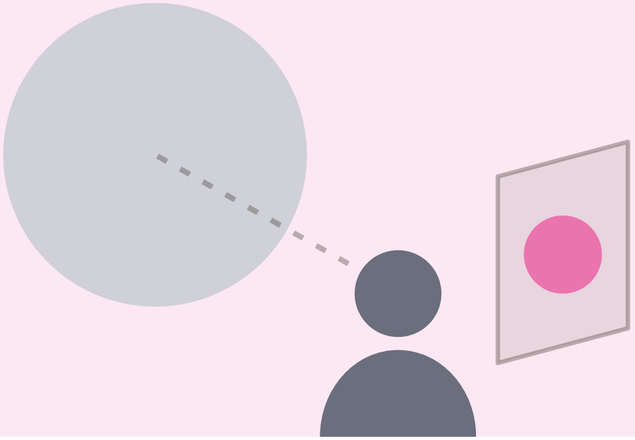


Camera space

Camera · embedded — Prototype



Camera · separated — Prototype

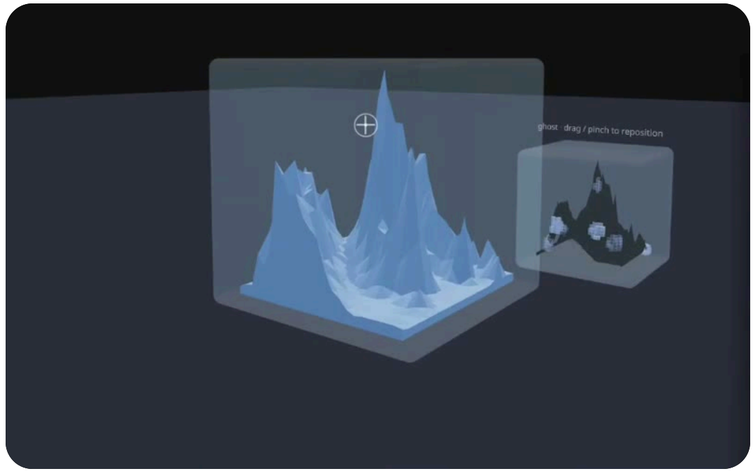


Always on
risks Midas touch

Explicit
you ask to see it

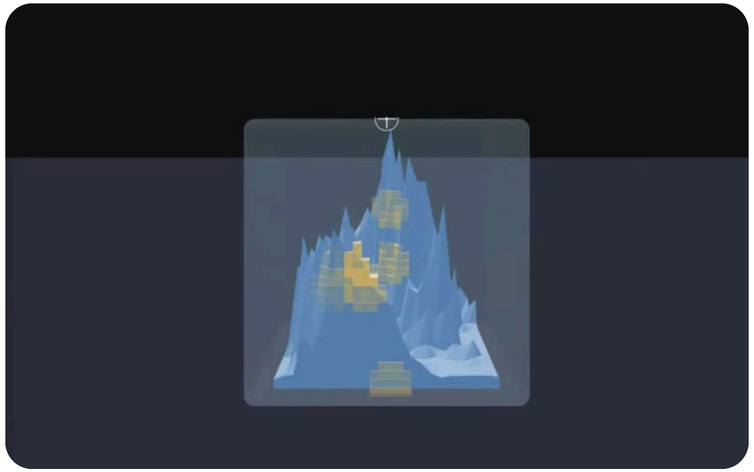
Implicit
the system decides

The other three quadrants



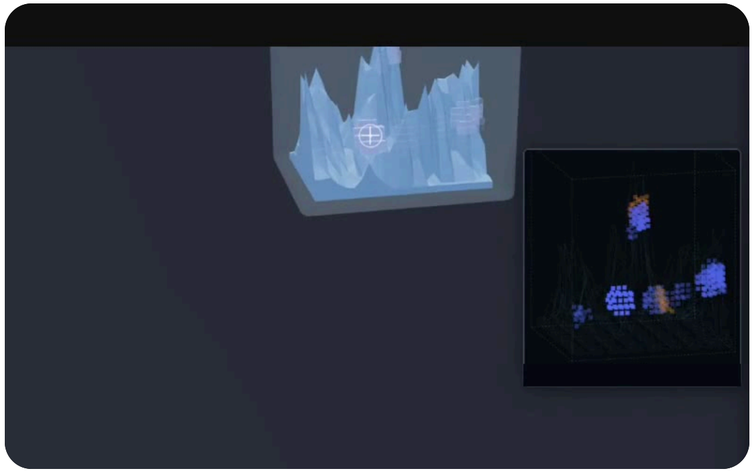
World, separated

A movable miniature



Camera, embedded

Attention on the bounding box



Camera, separated

A 2D panel beside the data

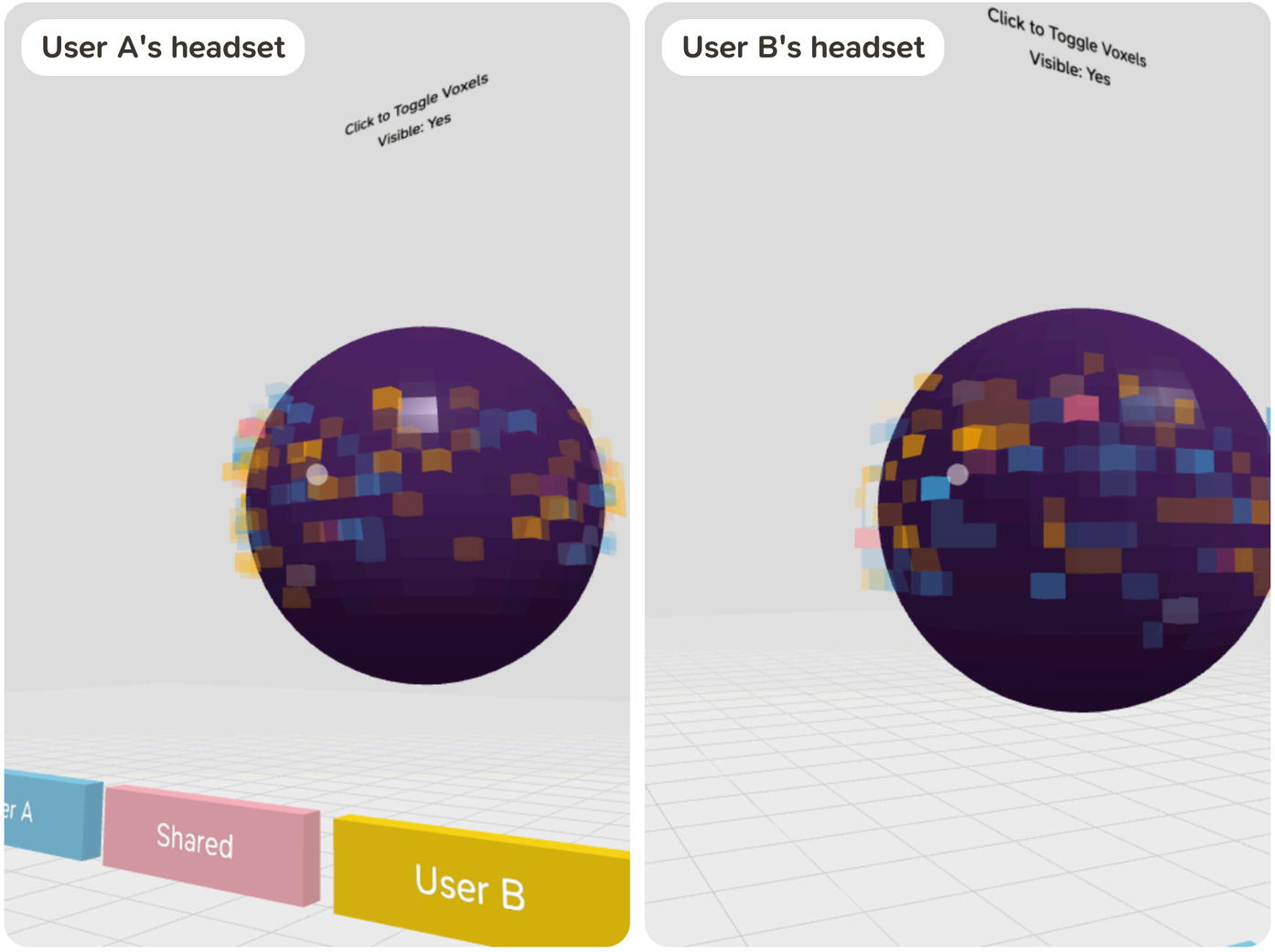
Proofs of concept, in VR only. Not part of the study.

3

Study

Does it help?

A treasure hunt, in the real system

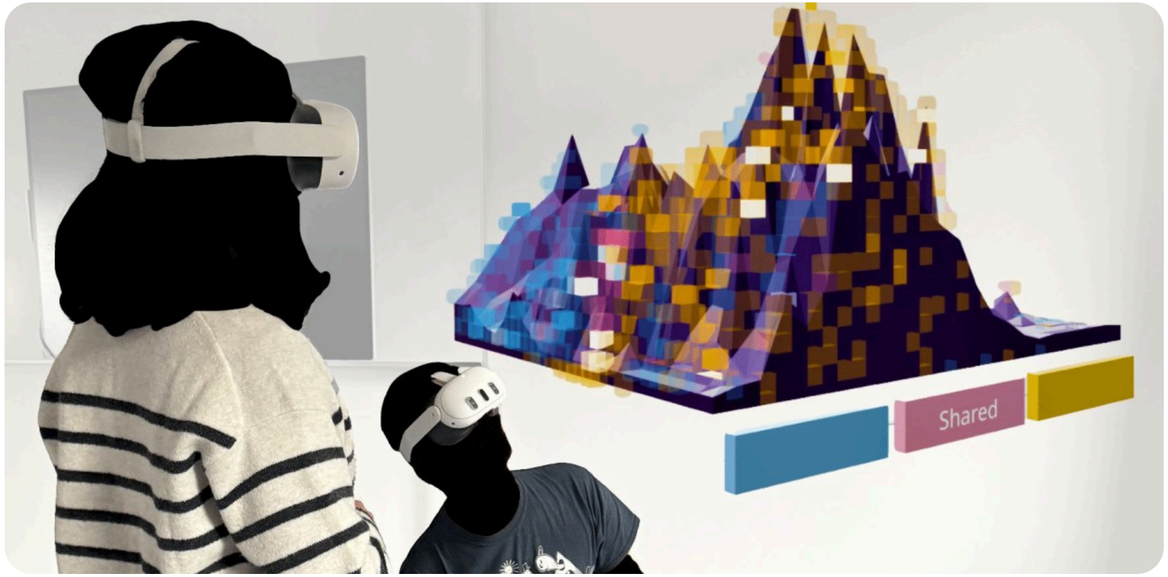


Live demo Drag a view to move that person · click the sphere to hide or show their voxels

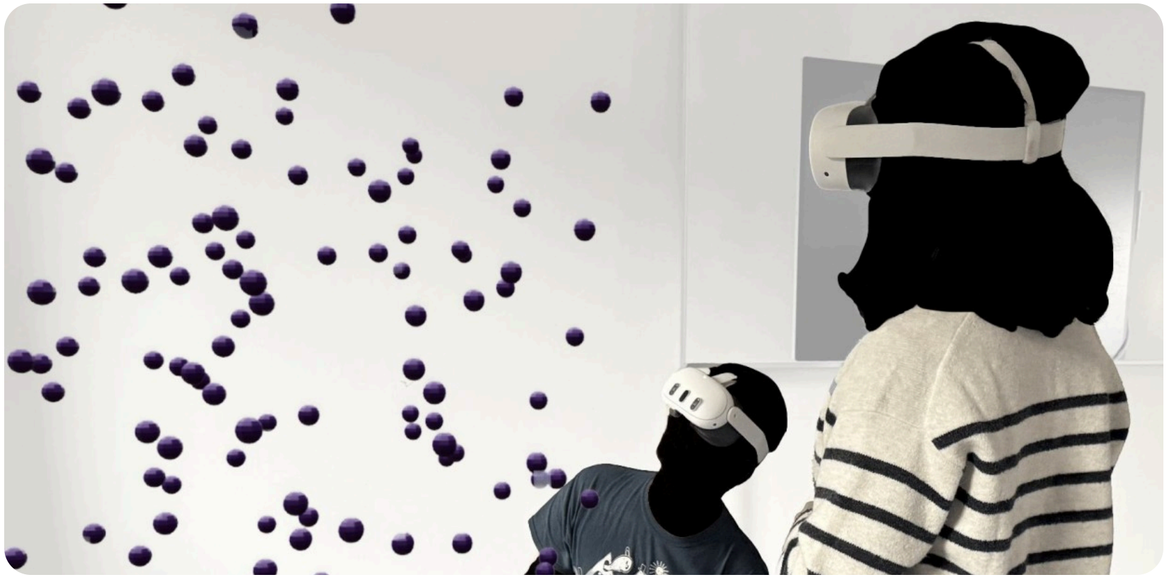
Reset

- **8 co-located pairs.** 16 participants, shared AR, 10-minute sessions.
- **Targets.** 5% of voxels.
- **Visible up close.** From the correct angle, within 10 cm.
- **With CAAV.** Voxels show where each person has looked.

A 2×2 mixed design



Terrain × CAAV



Scatter × No-CAAV

- **Within pairs.** Terrain or Scatter.
- **Between pairs.** CAAV or No-CAAV, four pairs per cell.

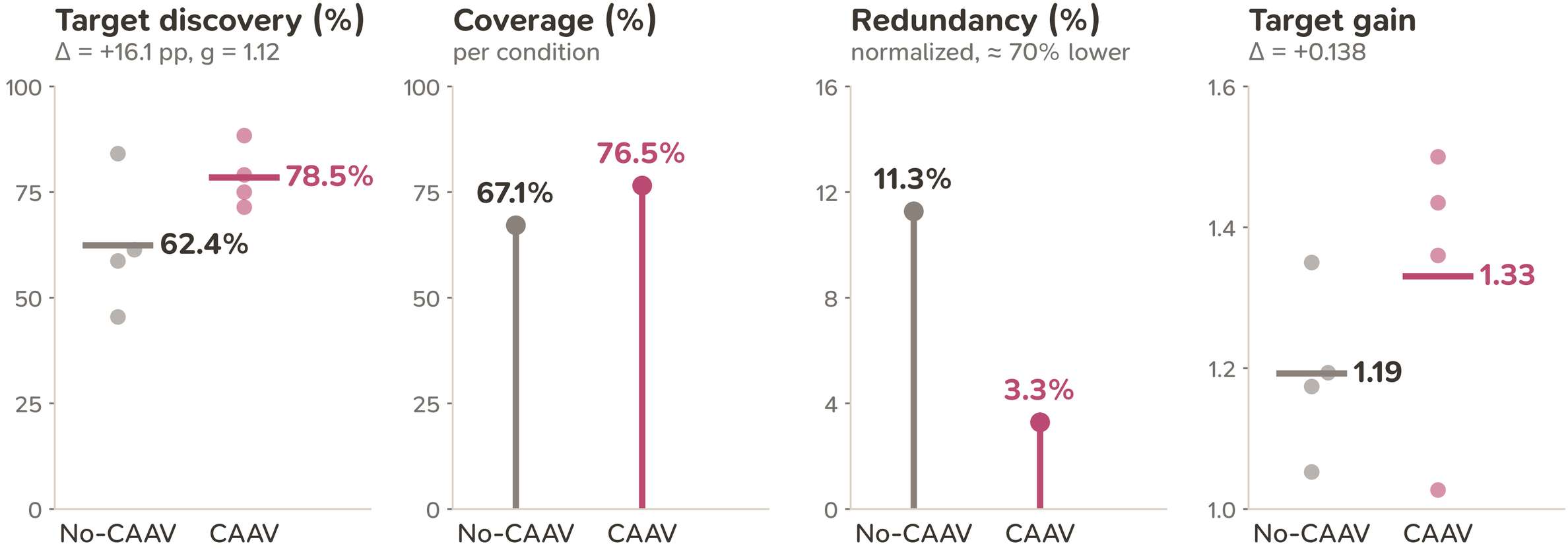
- **Target discovery, coverage.** Percentage of targets found, and of space explored.
- **Normalized redundancy.** $1 - H / H_{\max}$
- **Target gain.** $T_{\text{pair}} \,/\, \overline{T}_{\text{individual}}$

4

Results

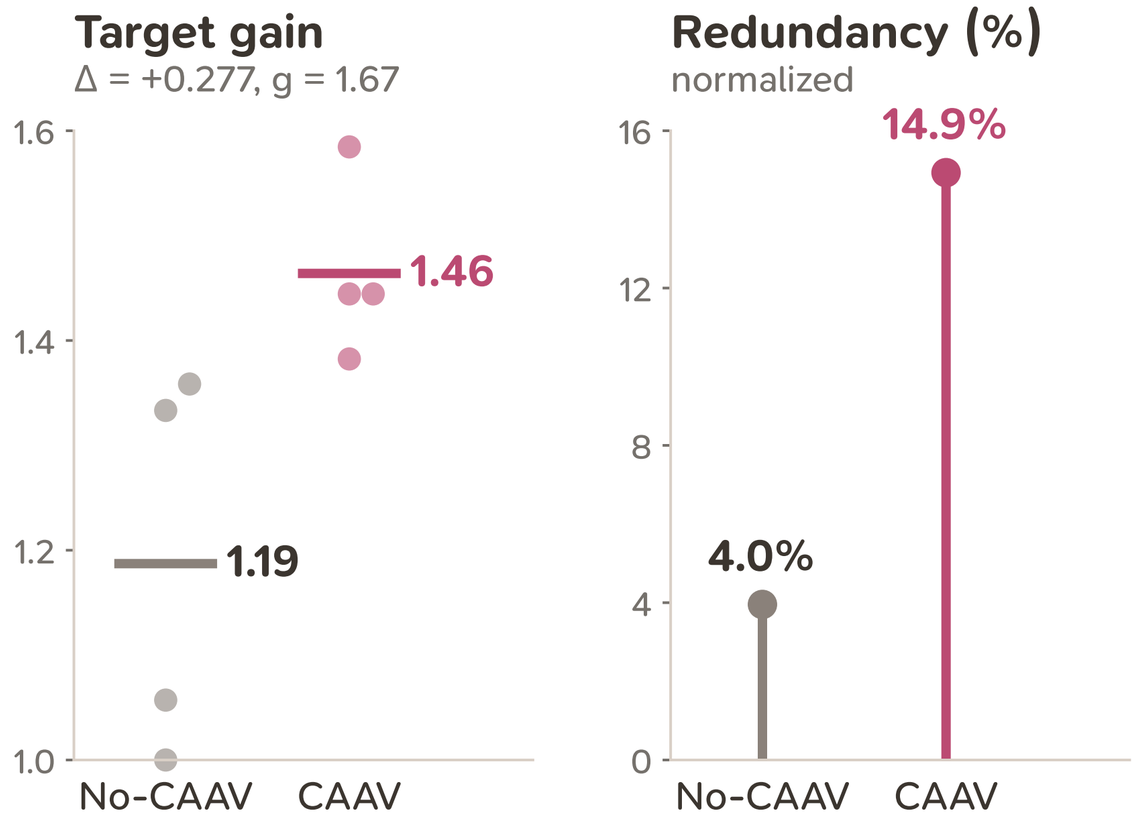
It depends on the space

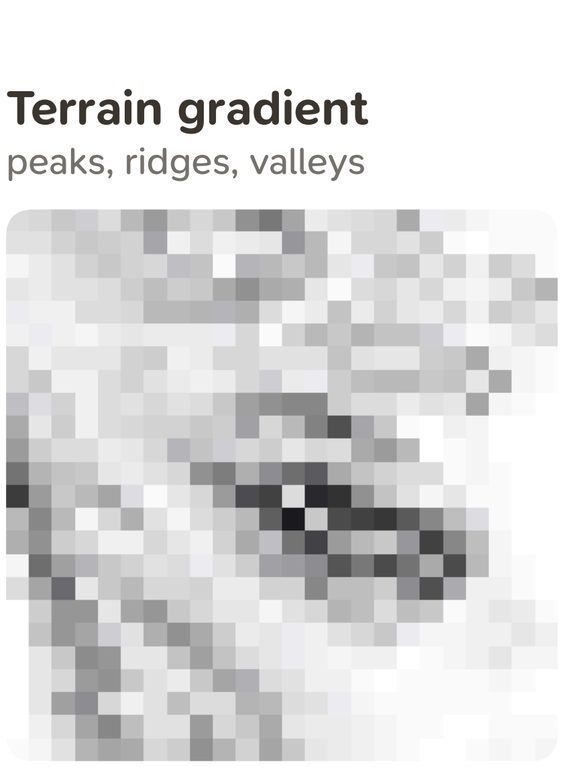
Scatterplot: less redundancy, more targets

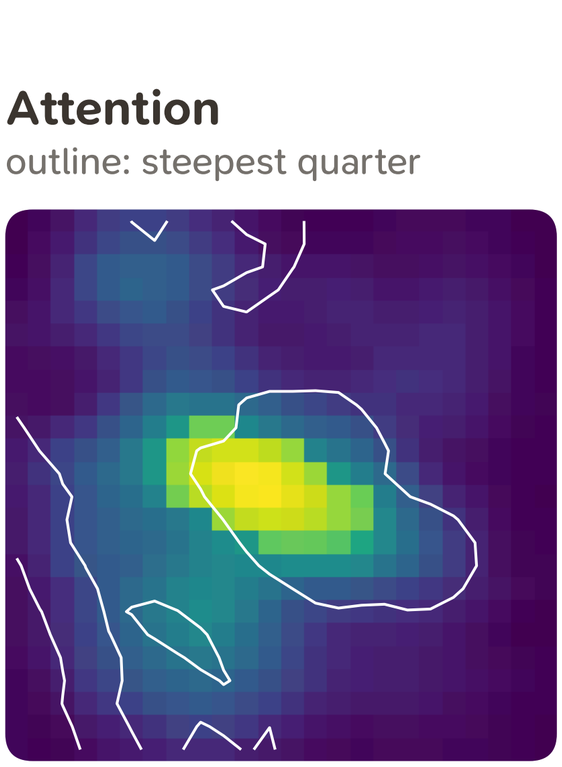


Dots: pairs. Bars: means. Targets BCa 95% CI [−0.012, +0.293]. Redundancy Δ = −0.058, exact p = 0.257: directional. Redrawn from Figs. 7, 8, 12.

Terrain: largest gain, more redundancy







Target gain BCa 95% CI [0.112, 0.450], $p = 0.029$. Gradient and attention: Pearson $r = 0.490$, Spearman $\rho = 0.561$, both $p < 0.0001$. Redrawn from Figs. 8, 9, 12.

Six strategies

Spatial division

We cut the space into top and bottom, and one search for top another search for the bottom.

P3

Systematic search

I took one half and systematically searched through it.

P12

Visual feedback

Used the voxel system to avoid redundant searches.

P14

Verbal coordination

We tried to coordinate verbally by describing positions relative to the Scatter structure.

P13

Landmark navigation

We used landmarks and described roughly in which area we found the targets.

P10

Physical synchronization

We placed the Scatter in the same location in the room using the yellow triangle.

P6

Top row, in pink: the strategies CAAV enabled.

By condition, and a clear preference

Strategy	Scatter No-CAAV	Scatter CAAV	Terrain No-CAAV	Terrain CAAV
Spatial division	2	4	3	2
Landmark navigation	2	0	3	5
Systematic search	0	4	1	2
Verbal coordination	3	0	1	1
Visual feedback	0	2	0	5
Physical synchronization	2	0	0	0

we have no verbal communication during the condition where we have [CAAV]

P8

93.8%

preferred CAAV conditions

24%

lower overall workload

NASA TLX, 2.79 vs. 3.69

74.38

mean SUS

benchmark 68

Participant counts per strategy, by pair. Participants may appear under more than one strategy.

Limitations and future work

Limitations

- **Scale.** Voxels get expensive at high resolutions or with many users.
- **Proxy.** Head orientation is coarser than eye tracking.
- **Scope.** Synchronous, co-located pairs, in visual search.
- **Baseline.** Only no attention visualization.

Future work

- **Tasks and teams.** Diverse tasks, larger teams, selective sharing.
- **Baselines.** Shared gaze rays, pointer trails, manual markers, minimaps.
- **Prototypes.** The other three quadrants, in networked AR.
- **Semantic attention.** Recording at the level of data elements.

Takeaways

- **CAAV effectiveness depends critically on visualization context.**
- **Scatterplots.** Reduced redundant exploration, and emergent coordination strategies.
- **Terrain.** A large coordination gain, even as attention visualization sometimes interfered with natural coordination cues.

Thank you for your attention



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